8.8 Product of Vectors

We have seen the notion of addition of two vectors, subtraction of one vector from another vector and the multiplication of a vector by a scalar. Now we study the notion of product of two vectors. There are two ways of multiplying two vectors.

(i) scalar product (dot product) and

(ii) vector product (cross product).

To define such products we need the angle between two vectors.

8.8.1 Angle between two vectors

Let $\vec{a}$ and $\vec{b}$ be any two vectors represented by $\overrightarrow{OA}$ and $\overrightarrow{OB}$ respectively. Angle between $\vec{a}$ and $\vec{b}$ is the angle between their directions when these directions are either both converge as in Fig. 8.36 or both diverge as in Fig. 8.34 from their point of intersection

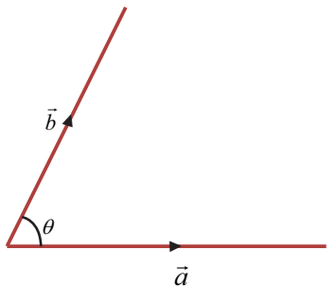


Fig. 8.34

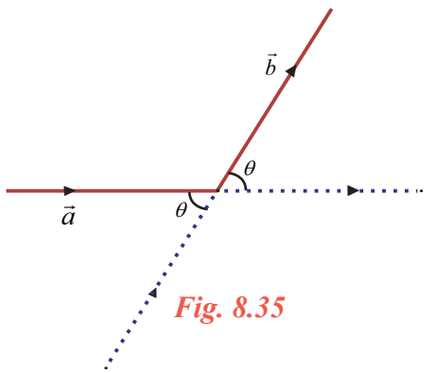


Fig. 8.35

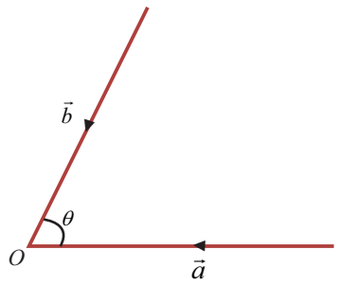


Fig. 8.36

Note that, if θ is the angle between two vectors then $0 \le \theta \le \pi$

When $\theta = 0$ or π, the vectors are parallel.

If two vectors neither converge nor diverge as in Fig. 8.35 then we can make them into either converge or diverge by extending the length of the vectors to find the angle between the two vectors.

8.8.2 Scalar product

Definition 8.16

Let $\vec{a}$ and $\vec{b}$ be any two non-zero vectors and θ be the included angle of the vectors as in Fig. 8.34.

Their **scalar product** or **dot product** is denoted by $\vec{a}\,.\,\vec{b}$ and is defined as a scalar $|\vec{a}||\vec{b}|\cos\theta$.

Thus $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta$.

Since the resultant of $\vec{a}\cdot\vec{b}$ is a scalar, it is called scalar product. Further we use the symbol dot (‘ · ’) and hence another name dot product.

Geometrical meaning of scalar product (projection of one vector on another vector)

Let $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$ and θ be the angle between $\vec{a}$ and $\vec{b}$.

Draw BL perpendicular to OA. From the right triangle OLB

$$\cos\theta = \frac{OL}{OB}$$

$$OL = OB\cos\theta = |\vec{b}|\cos\theta$$

But OL is the projection of $\vec{b}$ on $\vec{a}$

$$\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta = |\vec{a}|(OL)$$

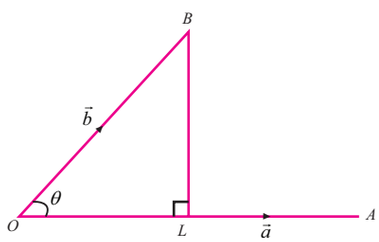


Fig. 8.37

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